HOMEWORK 1 --- REPORT • DLE-AI-202

Deep Learning

Neural-Network Foundations in PyTorch · Weeks 1--2

AI Academy · National Artificial Intelligence Center

Name	*your full name*	**Student ID**	*as on Moodle (this seeds your run)*
Section / group	*e.g. M504*	**Date**	*submission date*
Hours spent	*honest estimate*	**Submitted with**	`src/`, `hw1_notebook.ipynb`, `figures/`

1 Part A: written problems

One box per problem. Typeset the maths or embed a clean scan with `\includegraphics`*. Resize a box with its* `[...\baselineskip]` *argument.*

A1 Shapes and parameter count [3]

(a) shapes per layer, (b) parameters, (c) multiply–adds per example

A2 Losses come from likelihood [4]

(a) Gaussian NLL → MSE; (b) the stable BCE identity and why it cannot overflow; (c) $\partial L/\partial z = \hat{y} - y$

A3 Backpropagation by hand [6]

(a) forward values; (b) graph and $\boldsymbol{\delta}^{(2)}, \partial L/\partial W^{(2)}, \partial L/\partial \mathbf{b}^{(2)}$ with factors named first; (c) $\boldsymbol{\delta}^{(1)}$ through the ReLU gate, $\partial L/\partial W^{(1)}$; (d) the unit with no gradient and the dying-ReLU consequence

A4 Activations and vanishing gradients [4]

(a) σ' and its maximum; (b) number of layers for $0.25^{L} < 10^{-6}$; (c) Swish and GELU derivatives

A5 Initialisation [4]

(a) variance derivation for He; (b) the two standard deviations for $784 \to 256$; (c) why not zeros; (d) expected initial loss and two bugs that inflate it

A6 Optimisers [4]

(a) two SGD and two momentum steps and your observation; (b) Adam's first update with bias correction and its size; (c) decoupled weight decay

A7 Regularisation [3]

(a) weight decay $= (1 - \eta\lambda)$ shrinkage; (b) $\mathbb{E}[a'] = a$ for inverted dropout; (c) batch norm train versus inference

A8 Debugging [2]

(i)–(iv): most likely cause and first check

2 Part B: code status

Paste the last line of `pytest` (for example "34 passed"), and note anything that does not pass and what you think is wrong.

3 Part E: experiments

Embed `figures/final_curves.png` in E4. Cite your own numbers.

E1 Depth and initialisation [4]

E2 Optimisers [3]

E3 Regularisation [4]

include your results table (configuration, train acc, val loss min/last, test acc)

E4 Final model [4]

configuration and why (validation numbers), final test accuracy (evaluated once), where the errors are

4 Part C: reflection and disclosure

C1 Trace a gradient [4]

C2 Lines that must not be removed [3]

C3 Surprise, and AI-tool disclosure [3]

Which tool, for what? If none: “No AI tools used.”

Optional bonus

MyAdam / MixUp: what you did and what you observed